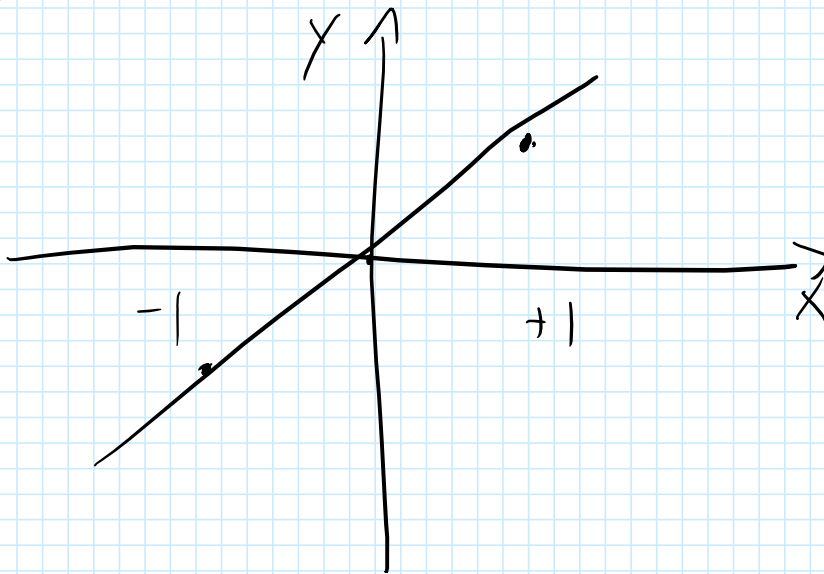


RETTA

$$Y = bX + C$$

$$\begin{matrix} x & y \\ (1, 2) \end{matrix}$$



IMPLICITA

$n \in \mathbb{N}, \mathbb{Z}, \mathbb{R}$

$$[ax + by + c = 0]$$

ESPLICITA

$$y = mx + q$$

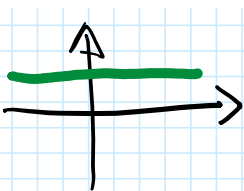
$$\frac{by}{b} = \frac{-ax}{b} - \frac{c}{b}$$

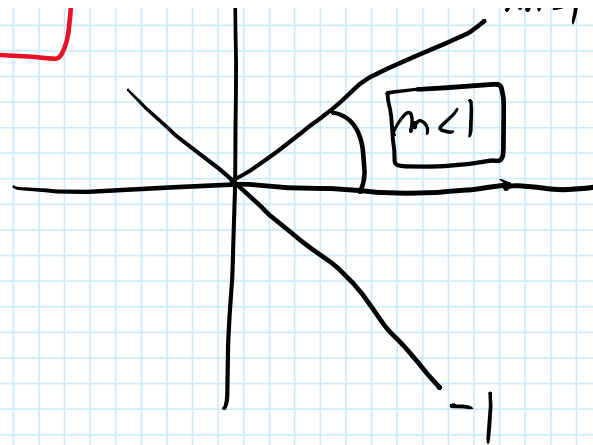
$$\begin{aligned} m &= \frac{-a}{b} \\ +q &= -\frac{c}{b} \end{aligned}$$

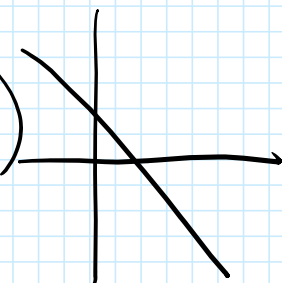
$$m > 0 \quad (y = 2x + 1)$$

$$m = 0 \quad (y = 2)$$

$$m = 1$$

$$m = 0 \quad (y = 3)$$


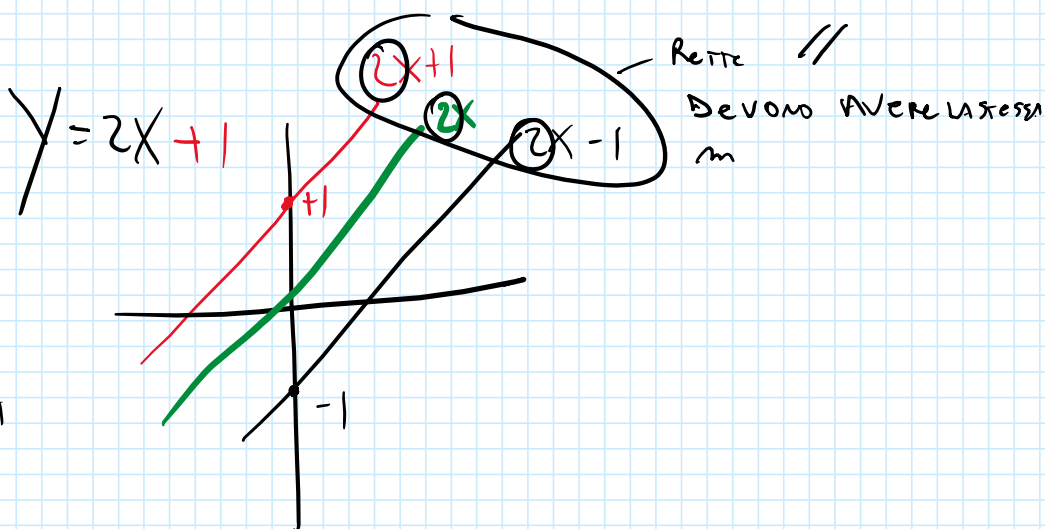


$$m < 0 \quad (y = -2x + 1)$$


$$q = 0 \quad (x = 0)$$

↓

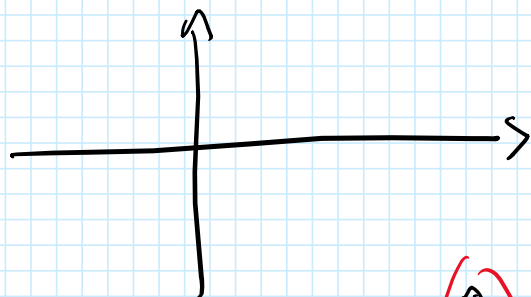
RETTE PASSA



RETTE VERTICALE

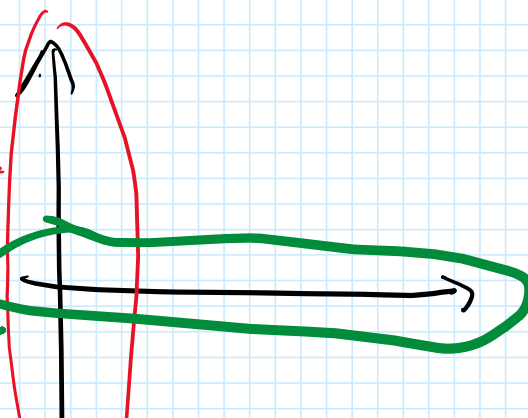
$$X = k_1$$

$$y = \text{IND}$$

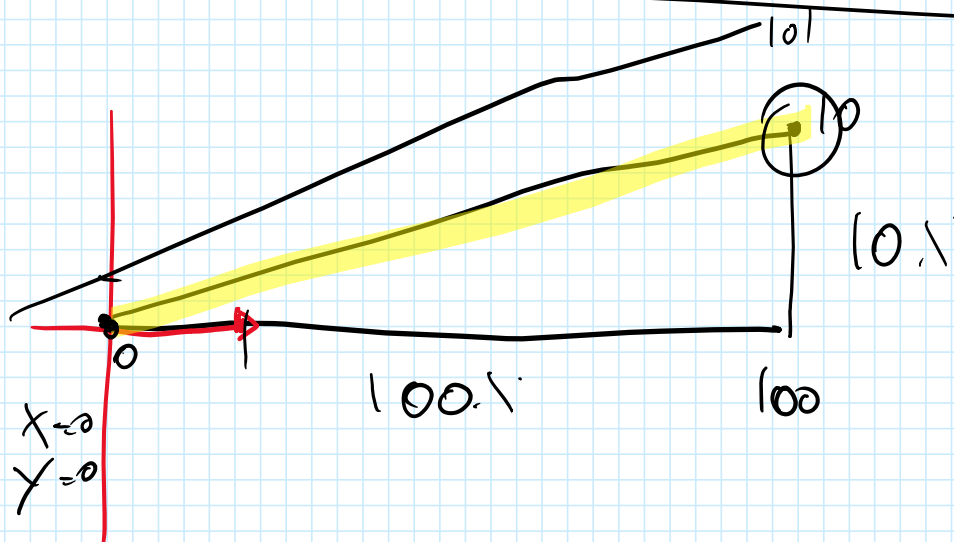


$$X = 0 \sim \text{Asse ORDINATE } (y)$$

$$Y = 0 \sim \text{Asse ASCISSE } (x)$$

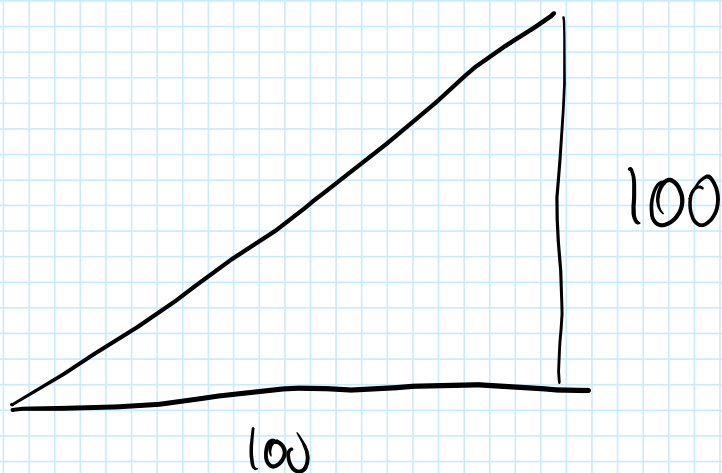


RETTA GENERICA $a \neq 0$
 $b \neq 0$



$$m = \frac{10}{100} = \frac{1}{10}$$

$$y = \frac{1}{10}x$$

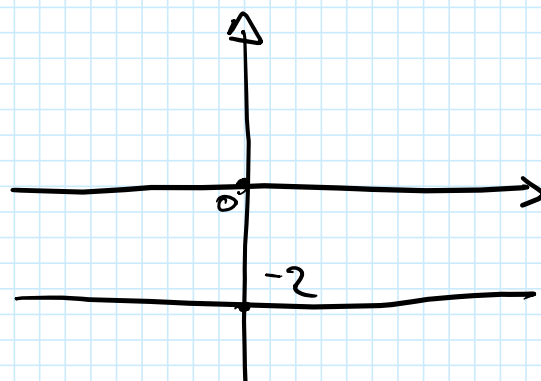


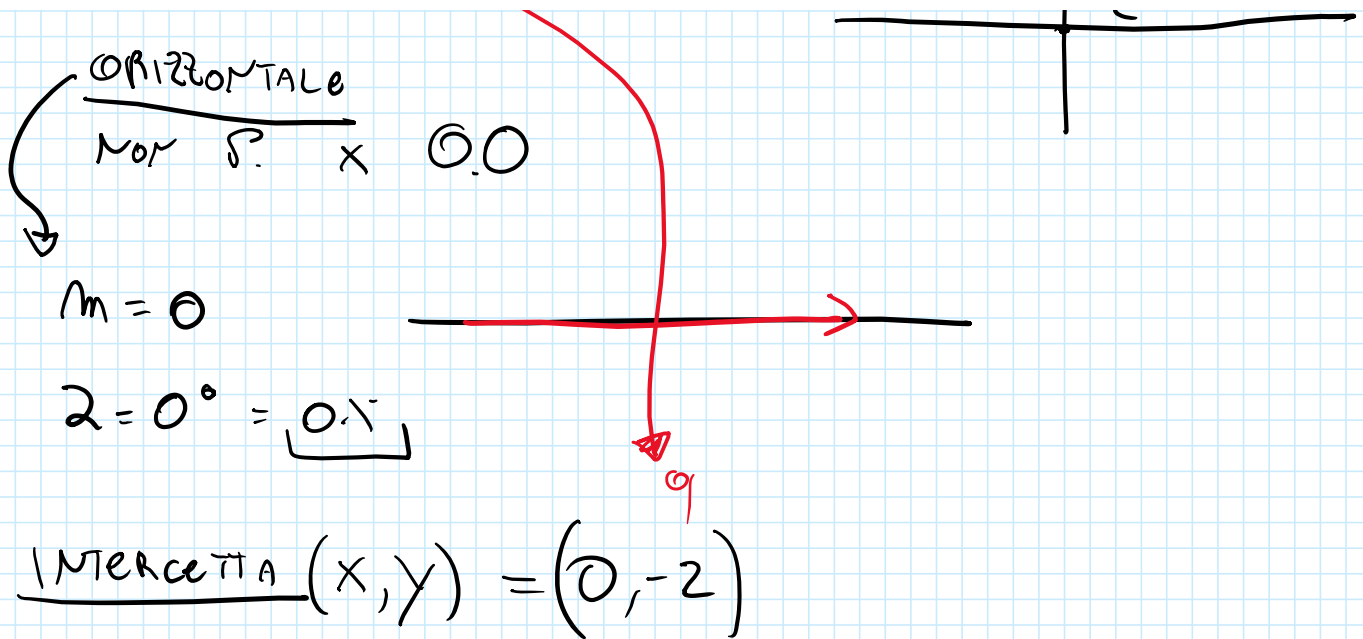
$$m = \frac{100}{100} = 1$$

2) $2y + 4 = 0$ IMPLICITA

$y = -2$ ESPLICITA

ORIZZONTALE





B) $y = -2x$ ESPlicita

$y = mx$

$+2x + y = 0$ IMPLICITA

x	y
0	0

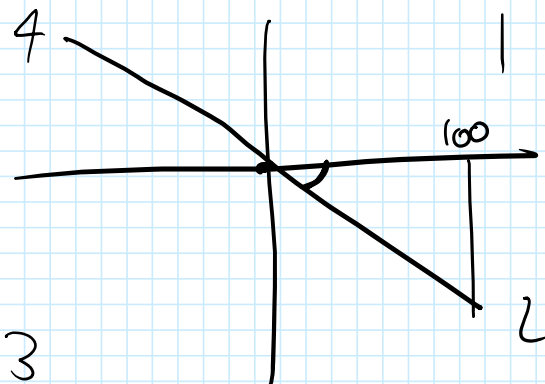
PASSA x 0.0

Generica

Coeff. ANGOLARE

$m = -2$

X	Y
100	-200



$$m = \frac{y}{x} = \frac{-200}{100}$$

$m = -2$

$-200x$

3

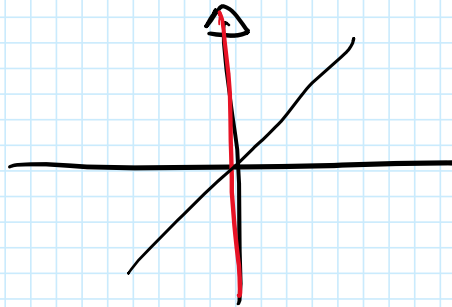
|

L

-200x

$$y = -2x$$

INTERCETTA



$$\begin{cases} X=0 \\ Y=-2X \end{cases}$$

ORD

$$\begin{cases} X=0 \\ Y=-2(0) \end{cases}$$

$$\begin{cases} X=0 \\ Y=0 \end{cases} \rightarrow \text{INTERCETTA}$$

(0,0)

$$5x - 20 = 0$$

$$5x = \frac{20}{5}$$

 $\forall y, x=4$

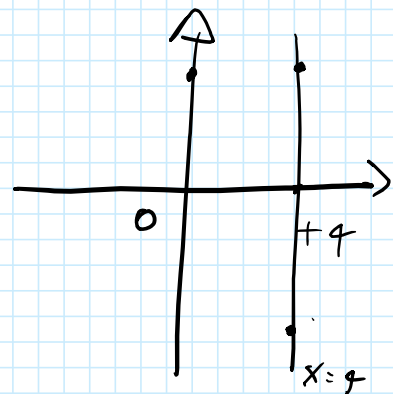
$$x = 4$$

VERTICALE

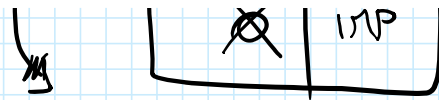
$$\text{IMPLICITA} \\ (ax + by + c = 0)$$

$$\text{ESPlicita} \\ (y = mx + q)$$

x	y
x	imp



VERTICALE



$$m=5$$

$$0y = 5x - 20$$

$$\text{Intercept} \begin{cases} X=0 \\ X=4 \end{cases} \text{ IMP}$$

d) $x + 3y - 6 = 0$ IMPLICITA

$$3y + x = 6$$

$$\frac{3y}{3} = \frac{-x + 6}{3}$$

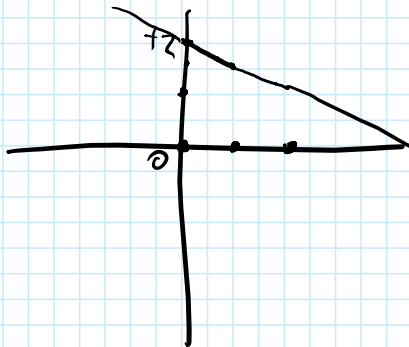
$$y = -\frac{1}{3}x + 2$$
 ESPlicita

X	Y
0	2
1	$\frac{5}{3}$
2	$\frac{4}{3}$

Coeff. ang

$$m = -\frac{1}{3}$$

$$\text{Intercept} (0, 2)$$



Pendenza decrescente

Pendenza decrecente



$$m = \frac{Y}{X} = \frac{-15}{100} = -0,15$$

$m \rightarrow$ Coef. Angolare = pendenza

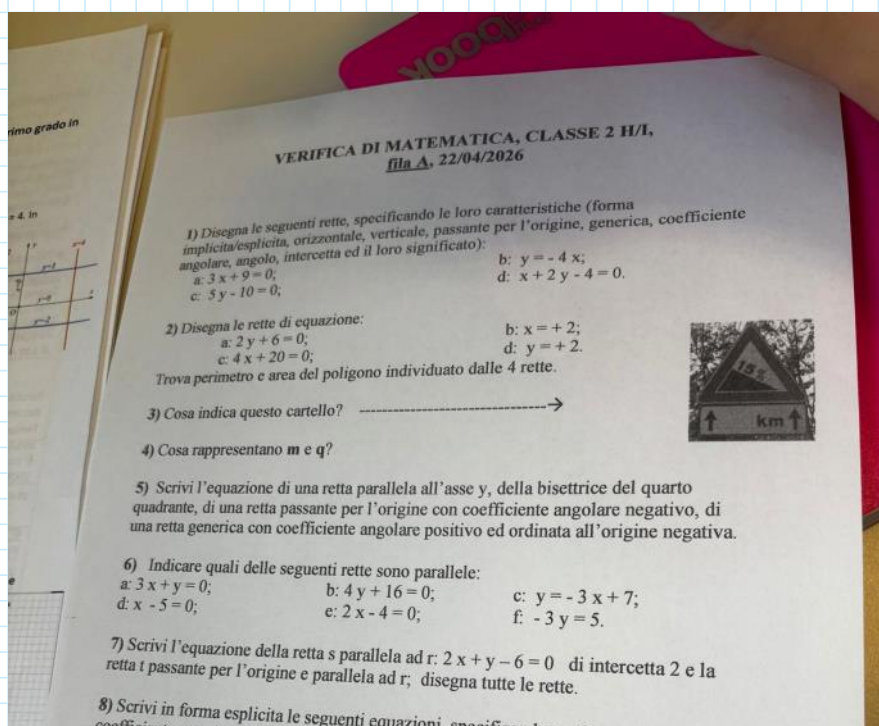
$\begin{cases} m > 0 \rightarrow \text{retta crescente} \\ m = 0 \rightarrow \text{orizz} \\ m < 0 \rightarrow \text{retta decrescente} \end{cases}$

Se due rette hanno la stessa m ($\bar{m}$), sono //

$q \rightarrow$ intercetta all'ordinata, punto di incontro con Asse y

Asse $x \rightarrow y = 0$

Asse $y \rightarrow x = 0$



7) Scrivi l'equazione della retta s parallela ad $r: 2x + y - 6 = 0$ di intercetta 2 e la retta t passante per l'origine e parallela ad r ; disegna tutte le rette.

8) Scrivi in forma esplicita le seguenti equazioni, specificando quali sono il coefficiente angolare m e l'ordinata all'origine q :

a) $5x + y - 2 = 0$;

b) $3x + 4y - 8 = 0$;

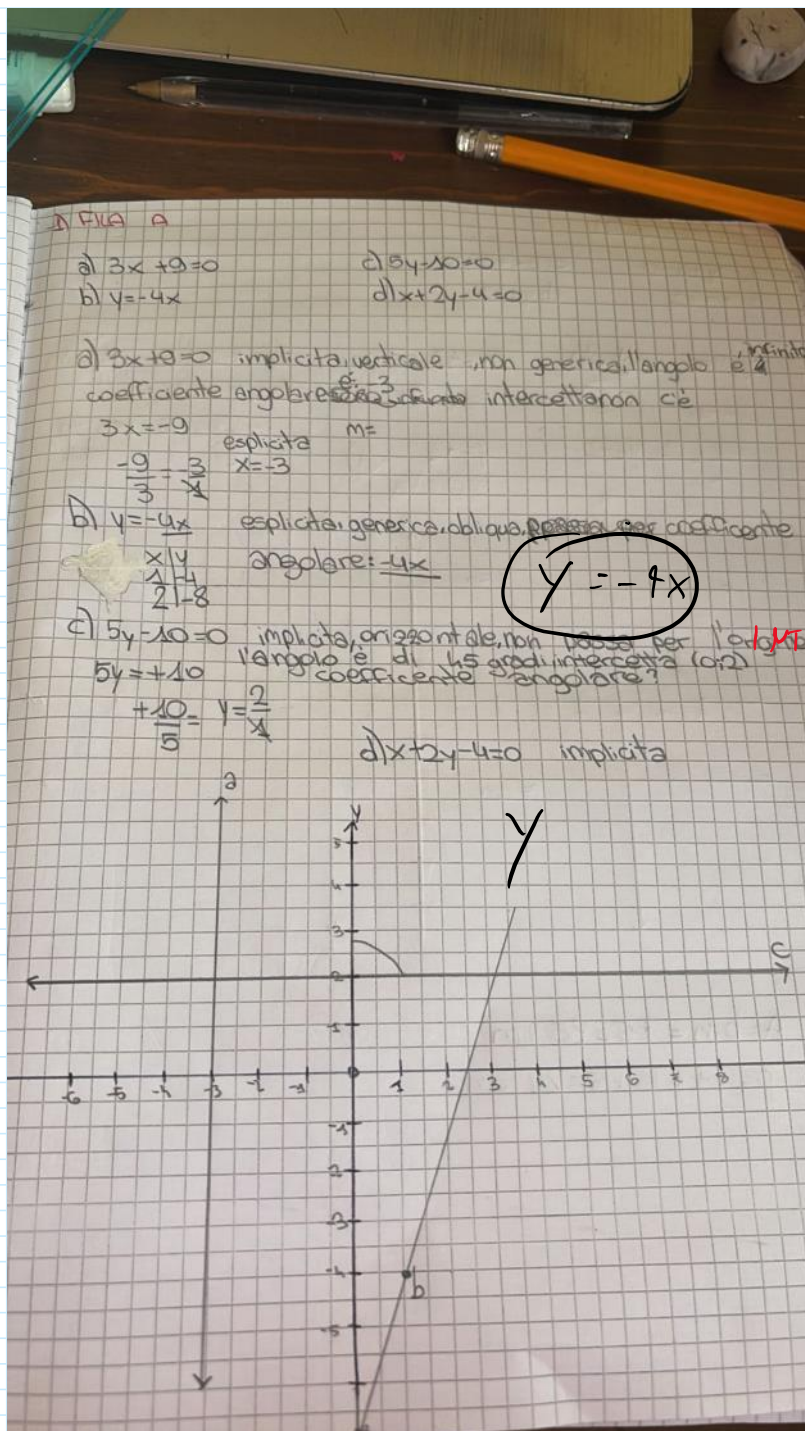
c) $7x + 3y = 0$;

d) $-y + 2 = 0$;

e) $8x = 24$;

9) Controllo quaderni

Esercizio	1	2	3	4	5	6	7	8	9	tot
Punteggio 10 +	14	10	8	6	6	10	10	10	16	100



$$y=$$

$$\{x=0\} \text{ (Asse } y)$$

x	y
0	0
2	-8

$$d: x+2y-4=0 \quad \text{in } \Gamma$$

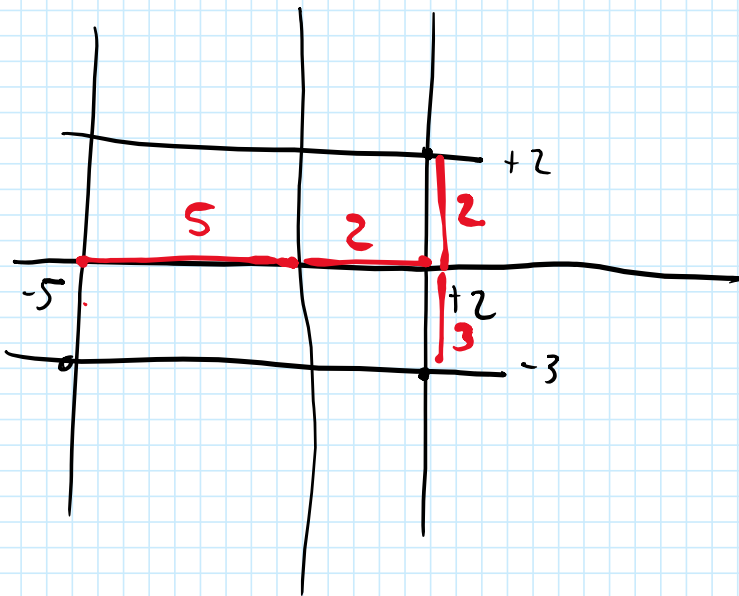
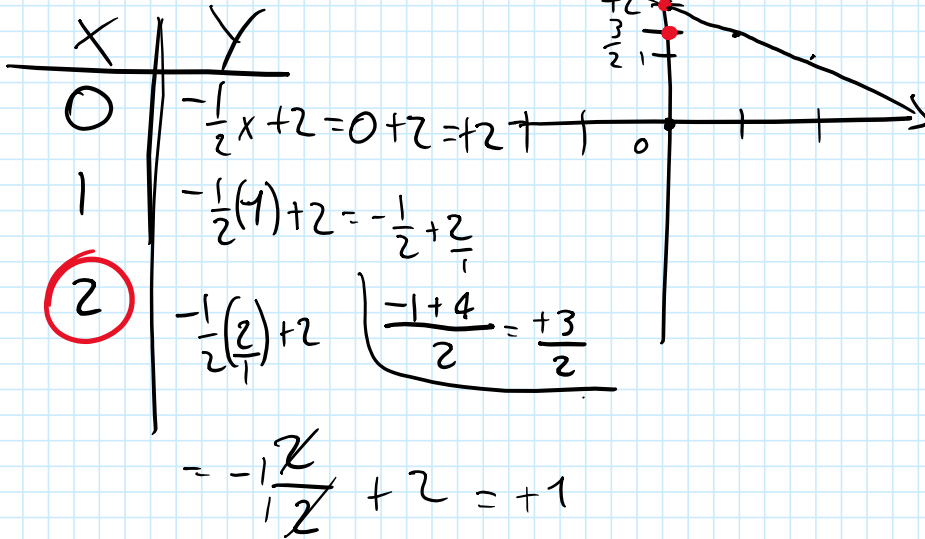
$$\frac{1}{2} \cdot y = \left[\frac{-x+4}{2} \right] = -\frac{1}{2}x + \frac{4}{2}$$

$$y = \left(-\frac{1}{2} \right) x + 2$$

$m < 0$

$$Y = \left(-\frac{1}{2}\right)X + 2$$

$$m < 0$$



$$2) \frac{2y}{2} = \left(\frac{-6}{2}\right)$$

$$\boxed{Y = -3}$$

esp

$$b) \boxed{X = +2}$$

$$o) \boxed{Y = +2} \text{ OR}$$

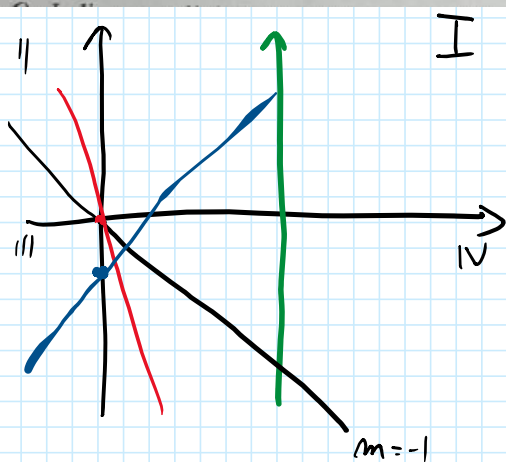
$$c) 4x + 20 = 0$$

$$1 \frac{4x}{4} = \underline{-20} \quad -5$$

$$1 \frac{x}{1} = -\frac{20}{4} - 5$$

$$x = -5 \quad \text{VERT}$$

5) Scrivi l'equazione di una retta parallela all'asse y, della bisettrice del quarto quadrante, di una retta passante per l'origine con coefficiente angolare negativo, di una retta generica con coefficiente angolare positivo ed ordinata all'origine negativa.



$$m > 0 \rightarrow \text{C.N.}$$

$$2: x = +3$$

$$b: y = -x$$

$$c: y = -2x$$

$$d: y = 3x - 1$$

$$2: 2x + 5y = 0 \quad b: 4y + 5 = 0 \quad c: y = -\frac{2}{5}x + 7$$

$$5y = -2x$$

$$y = -\frac{2}{5}x$$

$$4y = -5$$

$$y = -\frac{5}{4} \quad \text{OK! 2}$$

$$d: x - 3 = 0$$

$$x = 3$$

VERT

$$e: 2x - 7 = 0$$

$$\frac{2x}{2} = \frac{7}{2}$$

$$x = \frac{7}{2}$$

VERT

$$f: -2y = 5$$

$$y = -\frac{5}{2}$$

$$\bullet d \parallel c$$

$$\bullet b \parallel f \rightarrow m_b = m_f = 0$$

- $b // f \rightarrow m_b = m_f = 0$

- $2 // c \rightarrow m_2 = m_c = -\frac{2}{5}$

$$4x - y + 3 = 0$$

$$\frac{-y}{-1} = \frac{-4x - 3}{-1}$$

$$y = +4x + 3$$

$$8x - 4y + 2 = 0$$

$$\frac{-4y}{-4} = \frac{-8x - 2}{-4} =$$

$$y = +2x + \frac{1}{2}$$

$$6x - 2y + 4 = 0$$

$$\frac{-2y}{-2} = \frac{-6x - 4}{-2}$$

$$y = +3x + 2$$

$$3x + 5y - 7 = 0$$

$$\frac{5y}{5} = \frac{-3x + 7}{5}$$

$$\frac{y+3}{5} = \frac{-3x+1}{5}$$

$$y = \frac{-3}{5}x + \frac{1}{5}$$

$$x = n \rightarrow \text{VERT}$$

$$y = n \rightarrow \text{ORIZZ}$$

$$y = mx + q \rightarrow \text{Generica}$$

$m = \text{Coef. ANGOLARE}$

$\alpha = \arctan(m)$

$q = \text{INTERCETTA (ASSEY } x=0)$

$m \perp n \Rightarrow$
 $\alpha = 90^\circ$

$m=1$
 $\alpha = 45^\circ$